

SAPER: Soft Attention Pruner

Interpretability-Guided Soft Pruning of Attention Heads in Vision Transformers

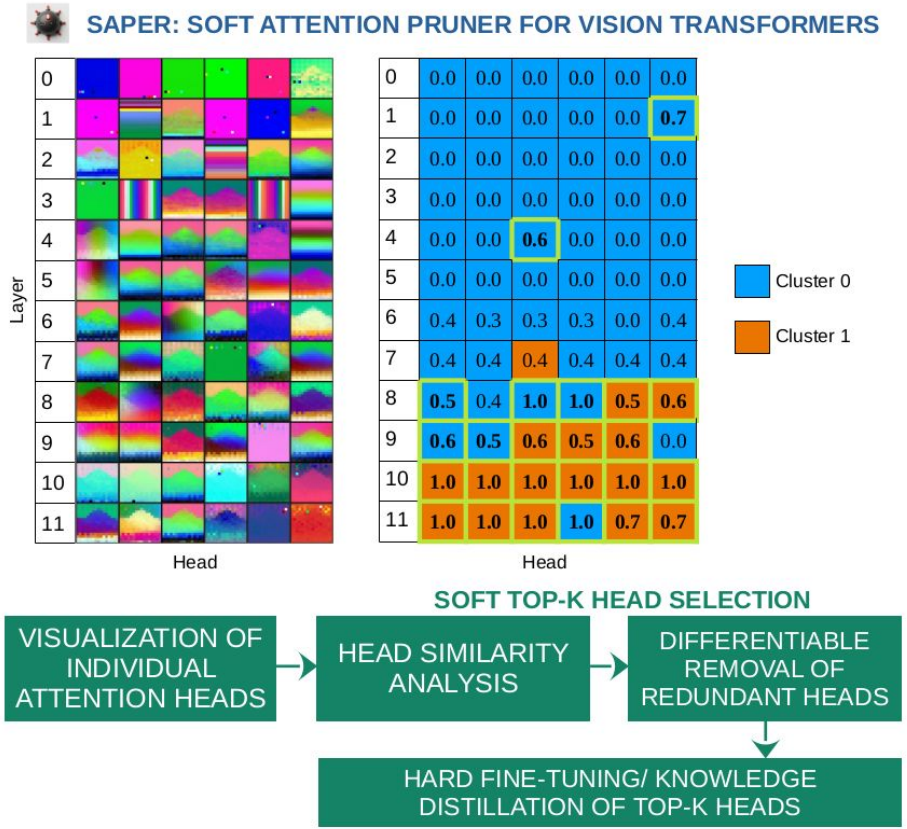
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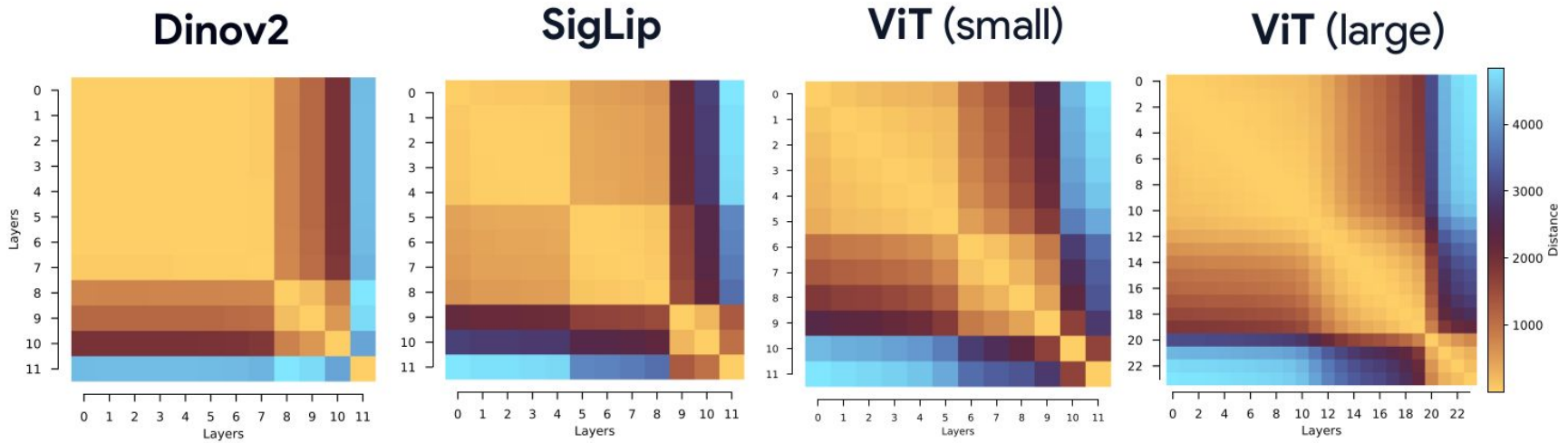
Centre for Credible AI, Warsaw University of Technology

Research Seminar

SAPER – brief description



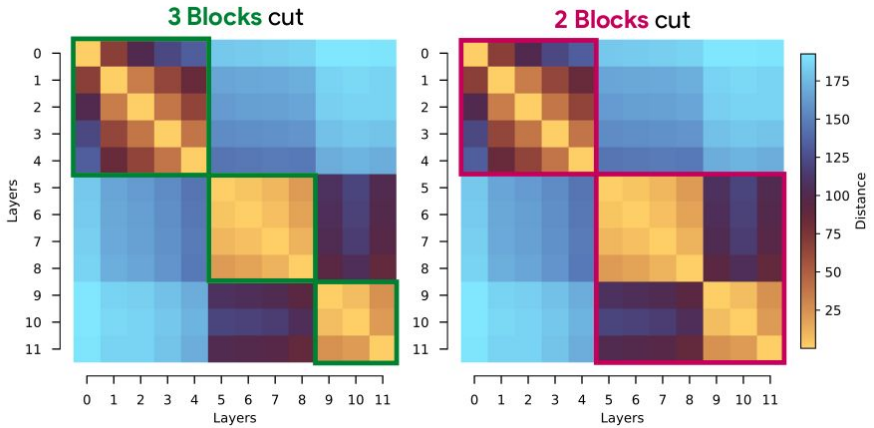
RAPTOR [1] – our predecessor



- Layer-layer similarity matrices across different vision transformers;
- **Contiguous block structure** along depth, phase-segmented regions of high similarity.

[1] M. Jacobs et al, Block-Recurrent Dynamics in ViTs, ICLR 2026.

RAPTOR – block-recurrent dynamics in ViTs



Max-cut segmentation of the layer-to-layer similarity matrix reveals contiguous recurrent phases in ViTs

- RAPTOR found representational similarity across layers.
- The authors proposed replacing the original L blocks using $k \ll L$ blocks applied recurrently.
- We propose **head-to-head** analysis (instead layer-to-layer).

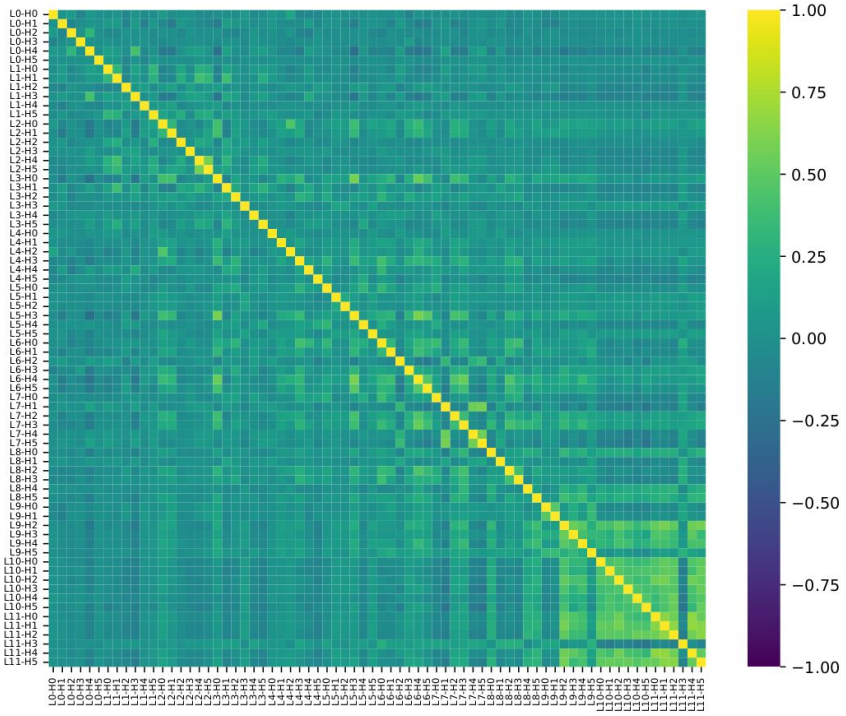
Semantic clustering

- We extract the pre-softmax attention logit matrix $A_i^h \in \mathbb{R}^{D \times D}$ for each attention head $h \in \{1, \ldots, H\}$ and image $i \in \{1, \ldots, N\}$, D is the total sequence length (i.e. the number of query and key tokens).
- We isolate the attention from the query CLS token to the key spatial tokens and apply softmax

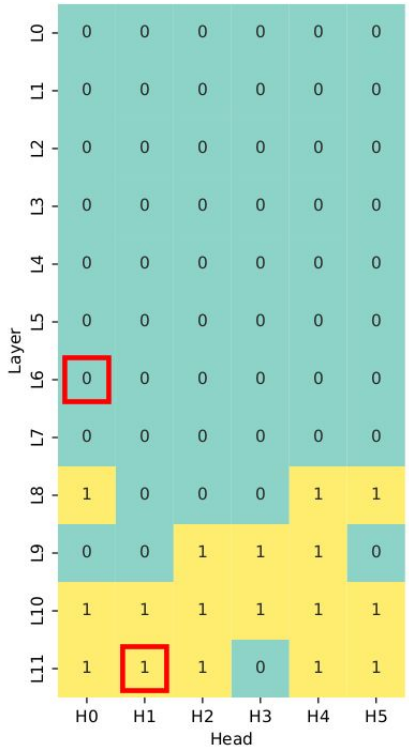
$$\mathbf{p}_i^h[j] = \frac{\exp(\mathbf{A}_i^h[0, j])}{\sum_{k=0}^{D-1} \exp(\mathbf{A}_i^h[0, k])}.$$

- We extract the **maximum attention value** (the most expressive patch) for each head, compute the Pearson correlation coefficient between all pairs of heads and apply the **spectral clustering**.

Semantic clustering: ImageNet-1K, ViT-S

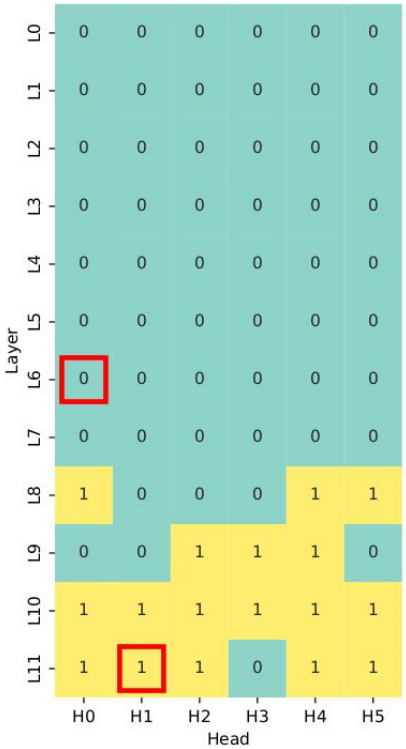


Head-to-head similarity matrix

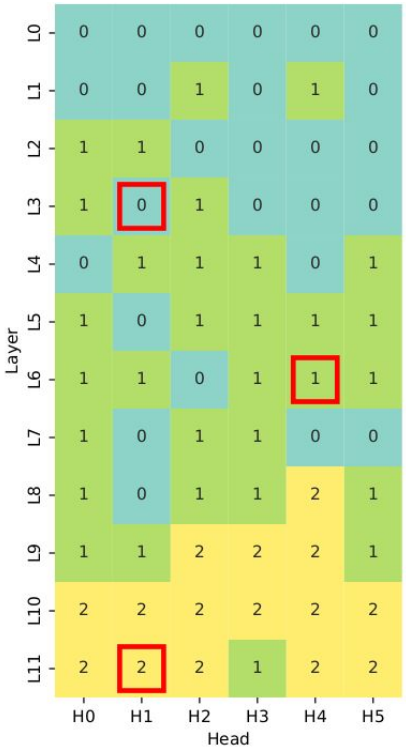


Spectral clustering, 2 clusters

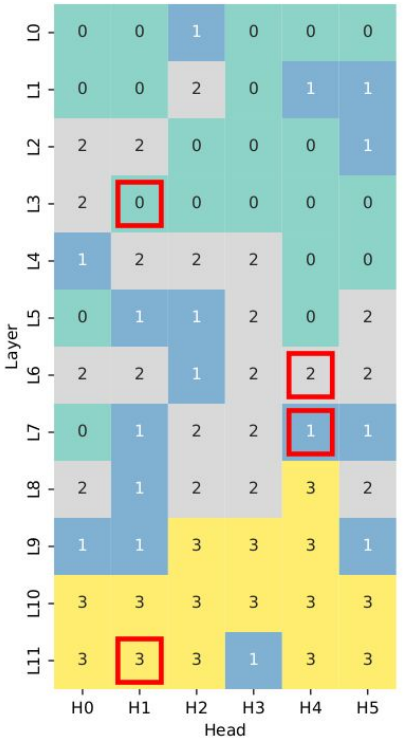
Semantic clustering: ImageNet-1K, ViT-S



2 clusters

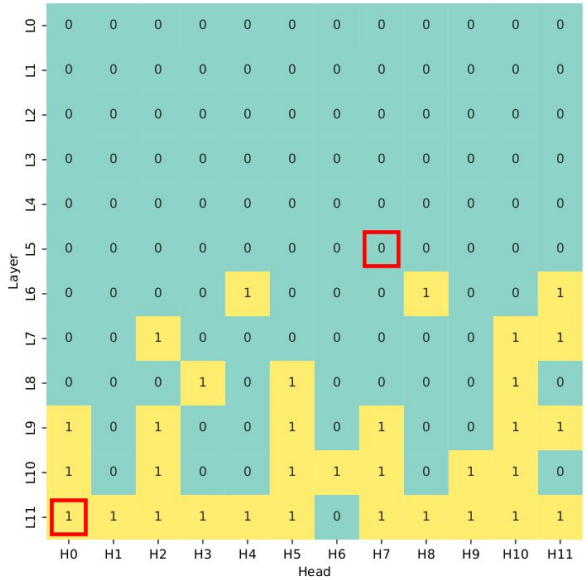


3 clusters

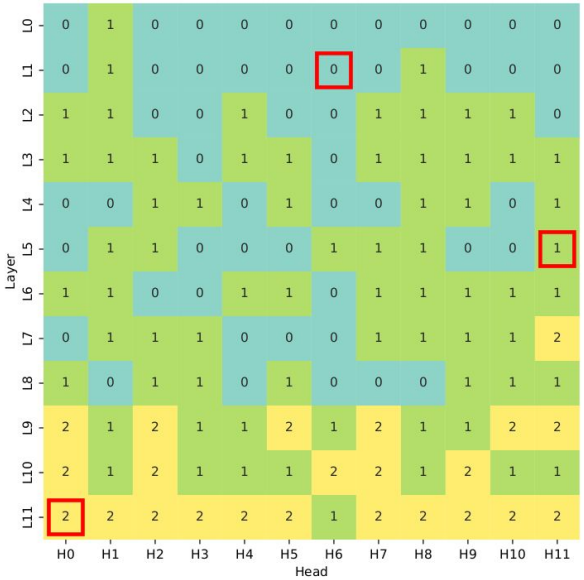


4 clusters

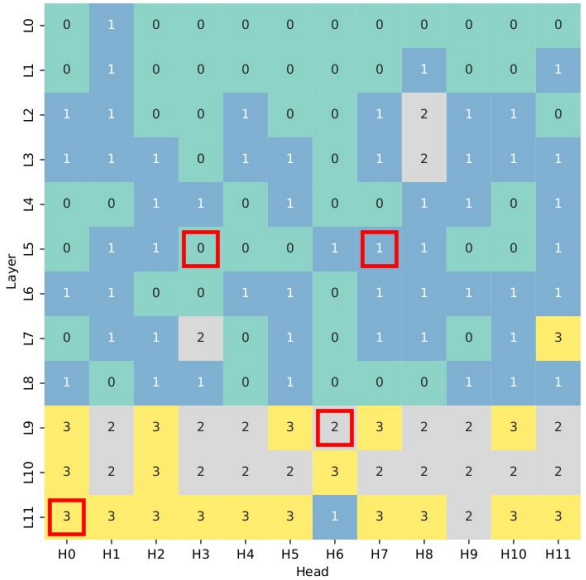
Semantic clustering: ImageNet-1K, ViT-B



2 clusters



3 clusters



4 clusters

Laplace maps

- An attention matrix can be interpreted as a weighted graph, where vertices represent tokens and edges encode the strength of the attention between pairs of tokens.
- We calculate a normalized Laplacian of the attention graph and extract the eigenvectors corresponding to the **second, third** and **fourth smallest eigenvalues**.
- Finally, we construct an RGB image mapping these eigenvectors to the red, green and blue channels.

Why do we select these eigenvalues?

➢ An eigenvector $\mathbf{x}$ of the Laplacian matrix L is related to the attention graph $\mathbf{A}$ through the Laplacian quadratic form:

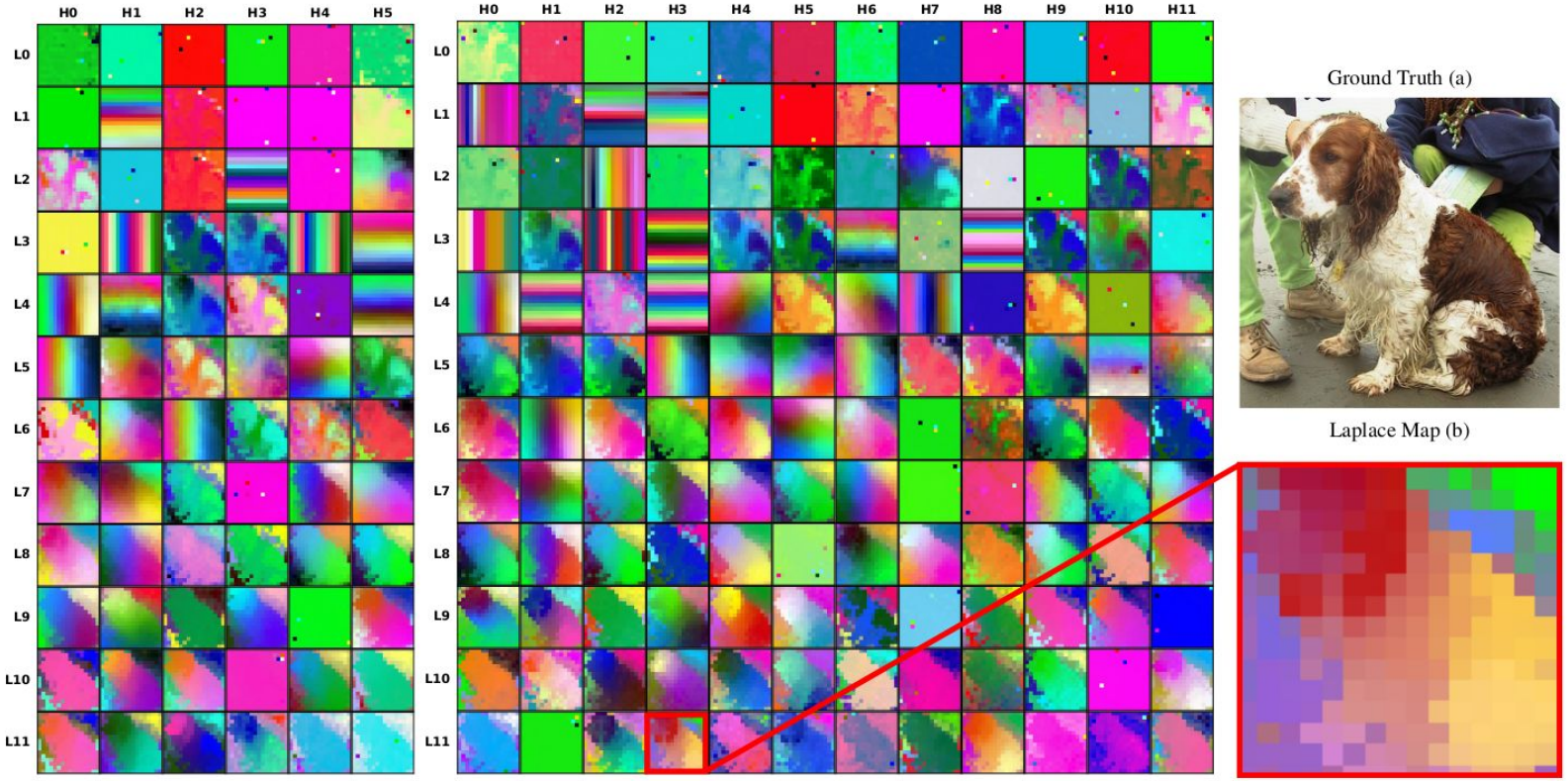
$$\mathbf{x}^\top L\mathbf{x} = \frac{1}{2}\sum_{i,j} A_{i,j}(x_i - x_j)^2$$

$A_{i,j}$ is the attention weight between token *i* and *j*, while x_i and x_j are the eigenvector values assigned to those tokens.

➢ Smaller eigenvalues correspond to minimizing the sum on the right!

➢ If two tokens have high attention weight, their eigenvector values must be very similar.

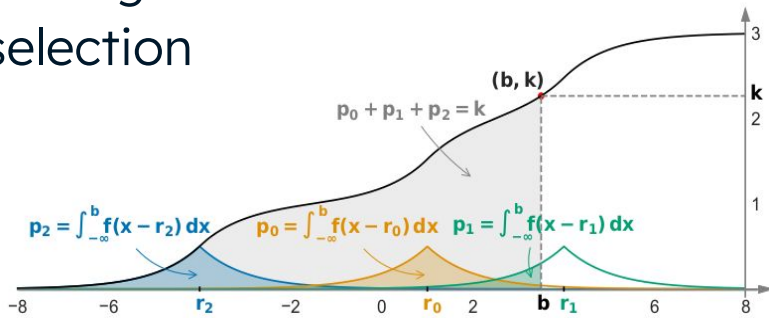
Laplace maps – examples



Differentiable attention head selection

- LapSum [2] offers a relaxation of the discrete Top-k selection problem, referred to as the Soft Top-k operation.
- We produce a score vector $r = (r_0,\ r_1,\ \ldots,\ r_{n-1}) \in \mathbb{R}^n$ and select an arbitrary continuous budget $k \in (0, n)$, producing the soft selection probabilities $p = (p_0, p_1,\ \ldots, p_{n-1})$:

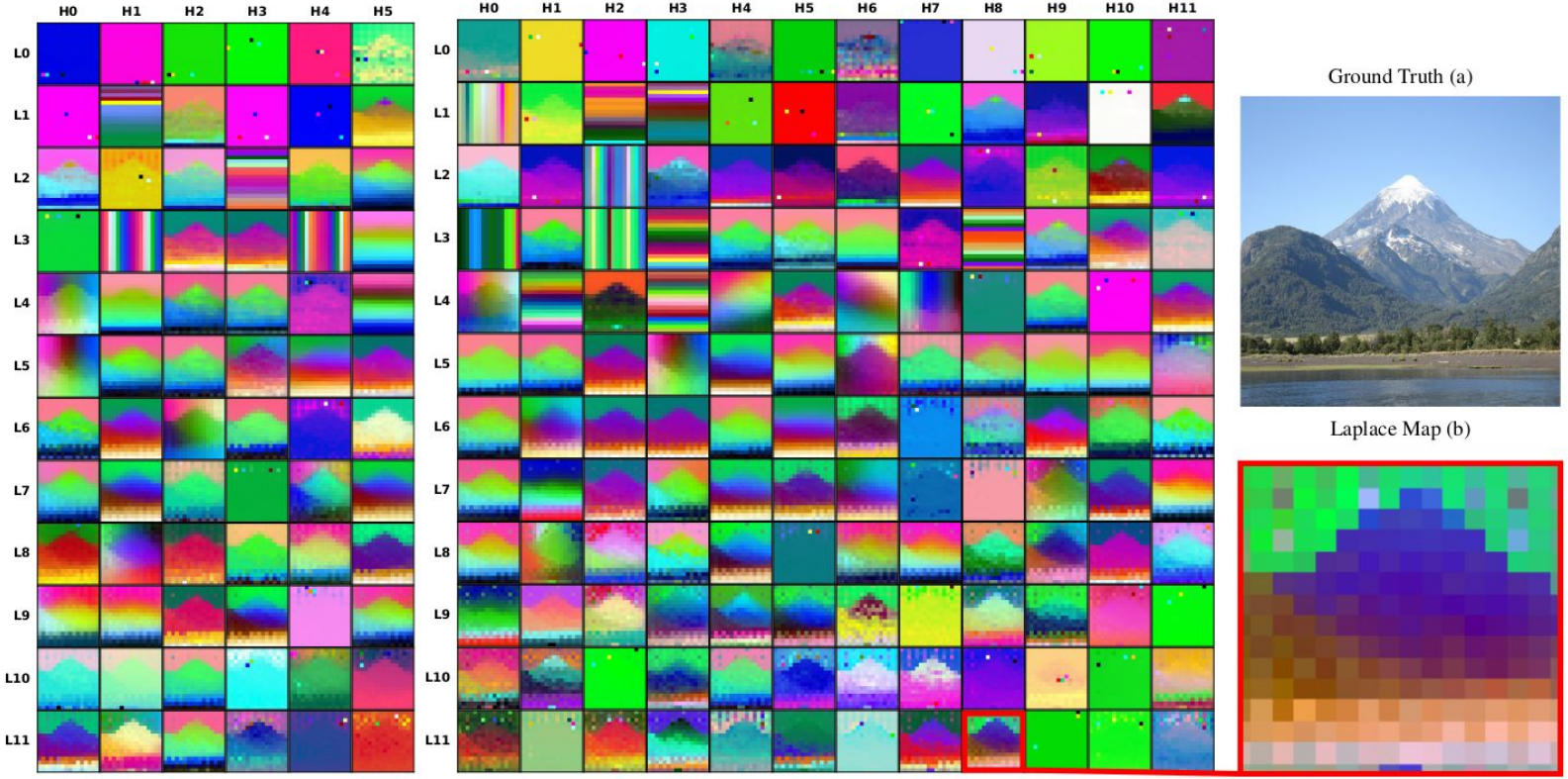
$$p_i = LapSum\,(r_i,\ \alpha) = Lap\left(\frac{b - r_i}{\alpha}\right)$$



$p_i \in [0, 1]$, $\sum_{i=0}^{n-1} p_i = k$, $Lap\,(\cdot)$ represents the CDF of the Laplace distribution, and α acts as a temperature. The time complexity is equal to $\mathcal{O}\,(n \cdot \log\,(n))$.

[2] Ł. Struski et al, LapSum – One Method to Differentiate Them All: Ranking, Sorting and Top-k Selection, ICML 2025.

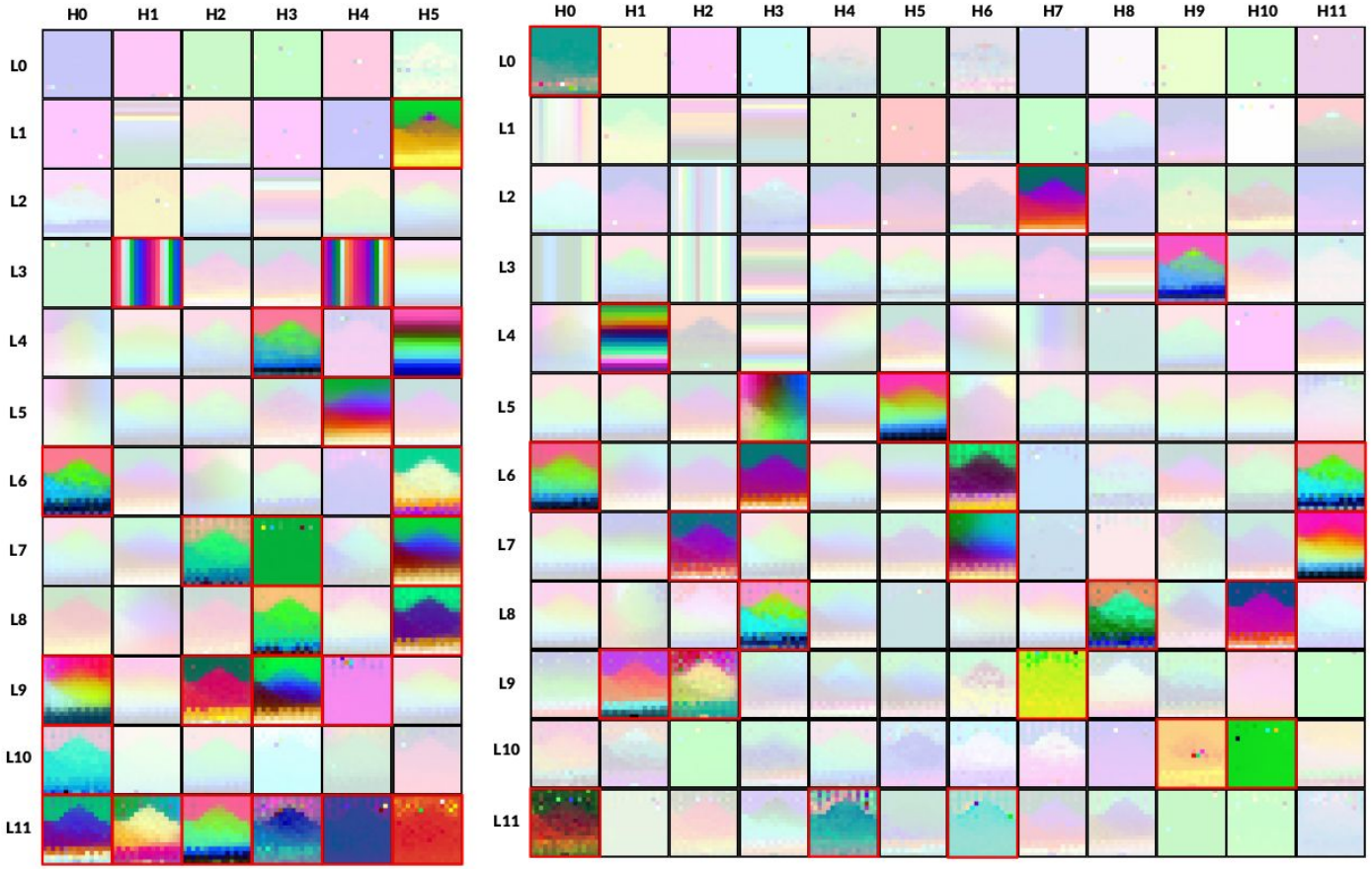
Laplace maps – examples



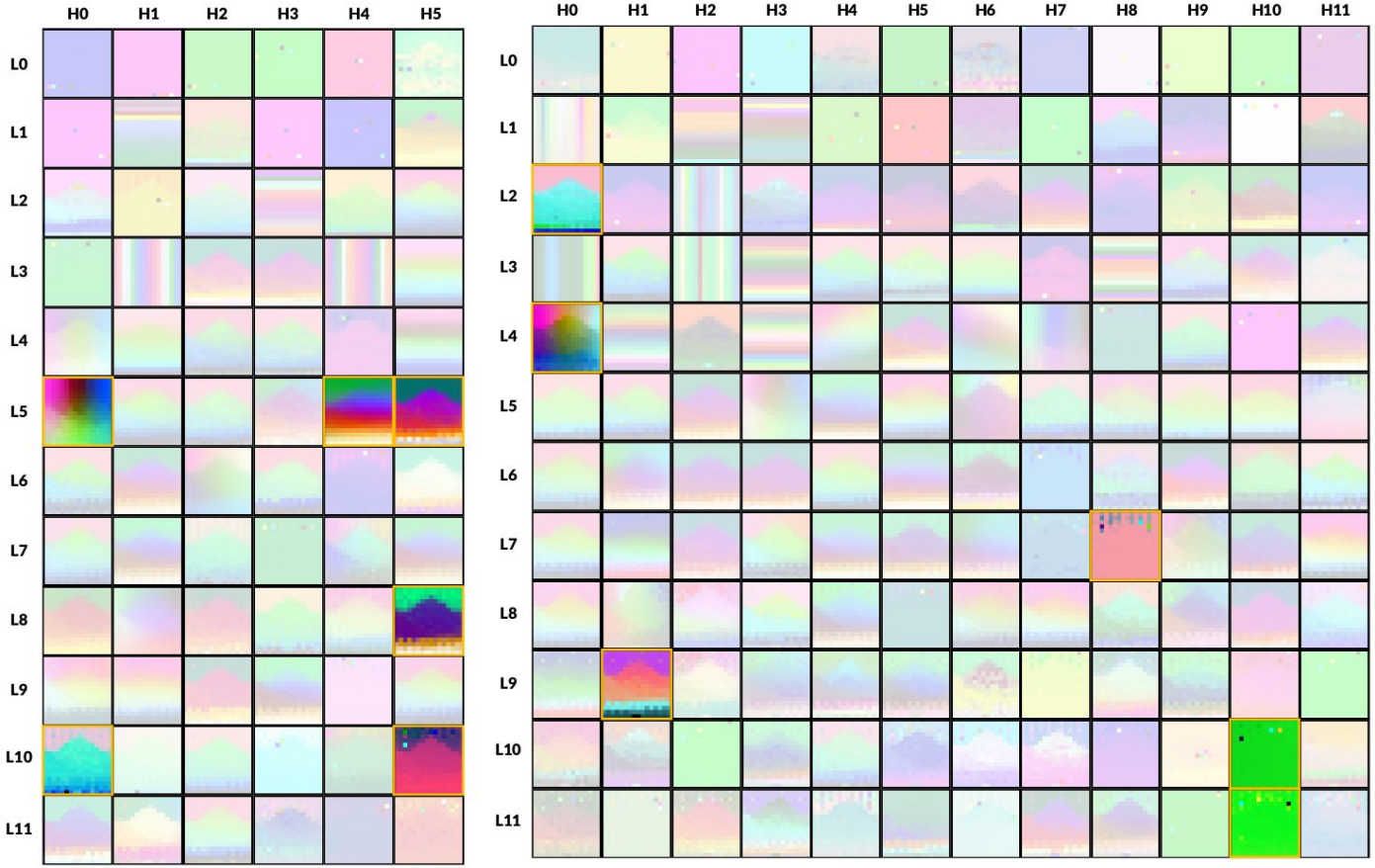
Main results

	ImageNet-1K				CIFAR-100			
	ViT-S		ViT-B		ViT-S		ViT-B	
	Fine-tuned	Distilled	Fine-tuned	Distilled	Fine-tuned	Distilled	Fine-tuned	Distilled
Method	**ACC**	**ACC**	**ACC**	**ACC**	**ACC**	**ACC**	**ACC**	**ACC**
DINOv2	$80.72_{\pm 0.21}$	N/A	$84.14_{\pm 0.32}$	N/A	$90.81_{\pm 0.10}$	N/A	$92.51_{\pm 0.49}$	N/A
RAPTOR_6	✗	✗	✗	✗	✗	✗	✗	✗
SAPER_6	$60.33_{\pm 0.33}$	$68.86_{\pm 1.39}$	$58.77_{\pm 2.27}$	$72.97_{\pm 1.78}$	$80.41_{\pm 0.85}$	$77.12_{\pm 0.19}$	$78.25_{\pm 0.55}$	$78.72_{\pm 0.13}$
RAPTOR_{12}	✗	✗	✗	✗	✗	$83.48_{\pm 0.52}$	✗	✗
SAPER_{12}	$68.18_{\pm 1.33}$	$71.78_{\pm 0.08}$	$64.33_{\pm 4.12}$	$77.54_{\pm 0.52}$	$\mathbf{84.48_{\pm 0.17}}$	$83.78_{\pm 0.58}$	$83.44_{\pm 0.44}$	$81.52_{\pm 0.66}$
RAPTOR_{24}	✗	–	✗	$\mathbf{81.2_{\pm 0.2}}$	✗	$85.74_{\pm 0.21}$	✗	$89.02_{\pm 0.22}$
SAPER_{24}	$74.82_{\pm 0.10}$	$77.99_{\pm 0.05}$	$74.83_{\pm 0.14}$	$80.58_{\pm 0.30}$	$\mathbf{87.57_{\pm 0.10}}$	$86.21_{\pm 0.05}$	$87.71_{\pm 0.34}$	$\mathbf{89.66_{\pm 0.69}}$
RAPTOR_{36}	✗	–	✗	$\mathbf{83.0_{\pm 0.1}}$	✗	$86.70_{\pm 0.32}$	✗	$\mathbf{90.46_{\pm 0.20}}$
SAPER_{36}	$77.47_{\pm 0.06}$	$78.94_{\pm 0.10}$	$77.68_{\pm 0.06}$	$81.32_{\pm 0.20}$	$\mathbf{88.90_{\pm 0.23}}$	$86.91_{\pm 0.19}$	$89.19_{\pm 0.33}$	$90.29_{\pm 0.08}$
RAPTOR_{48}	✗	–	✗	$\mathbf{83.2_{\pm 0.1}}$	✗	$87.34_{\pm 0.40}$	✗	$90.85_{\pm 0.10}$
SAPER_{48}	$79.17_{\pm 0.09}$	$79.39_{\pm 0.05}$	$79.67_{\pm 0.11}$	$81.99_{\pm 0.16}$	$\mathbf{89.79_{\pm 0.18}}$	$87.56_{\pm 0.21}$	$90.18_{\pm 0.20}$	$\mathbf{91.39_{\pm 0.13}}$

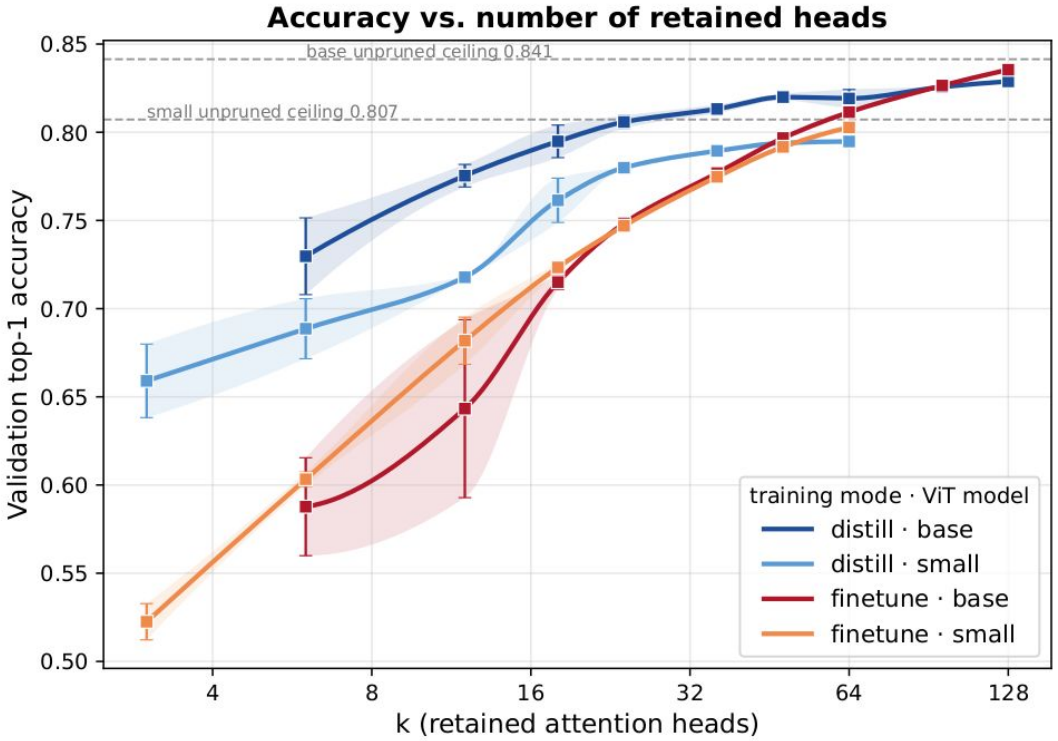
SAPER – heads retained after pruning, k=24



SAPER – heads retained after pruning, k=6



Head number ablation study



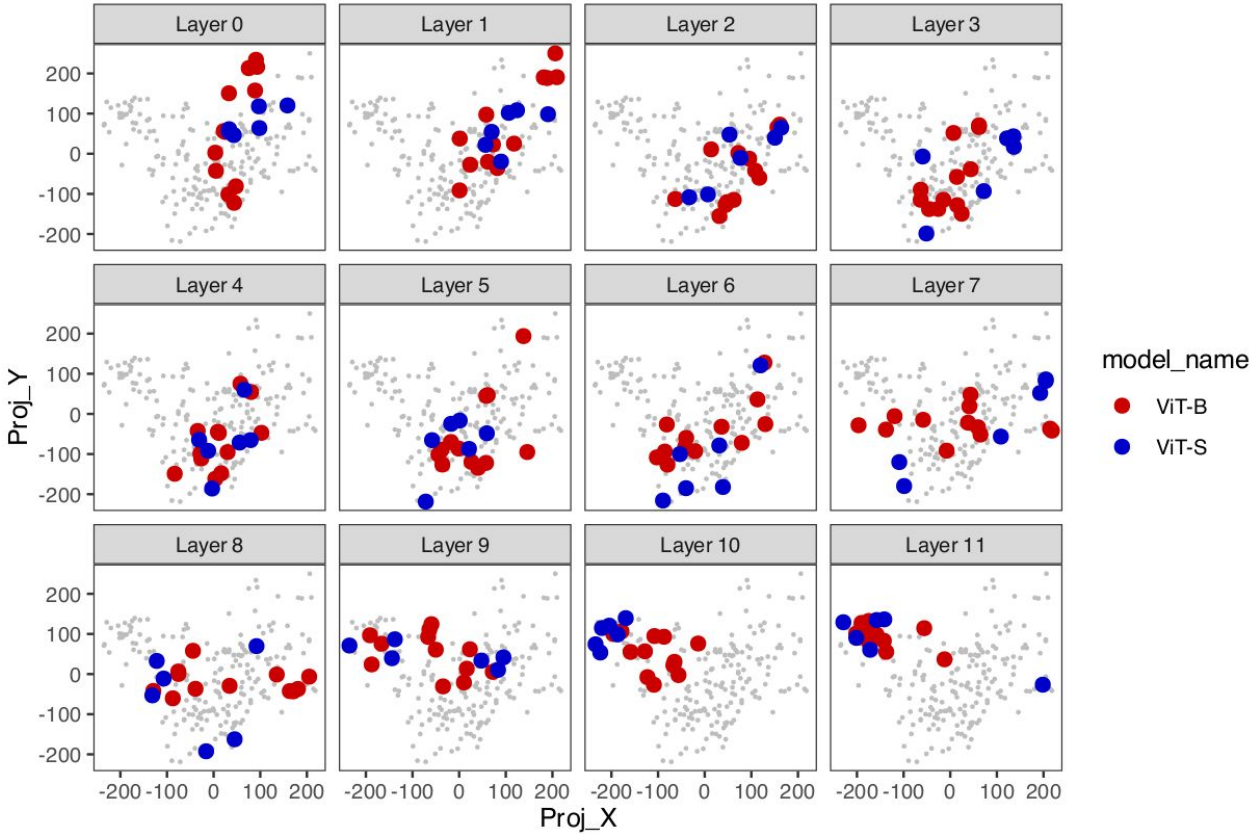
Results for ImageNet-1K, averaged over 2 seeds.

Principal component analysis

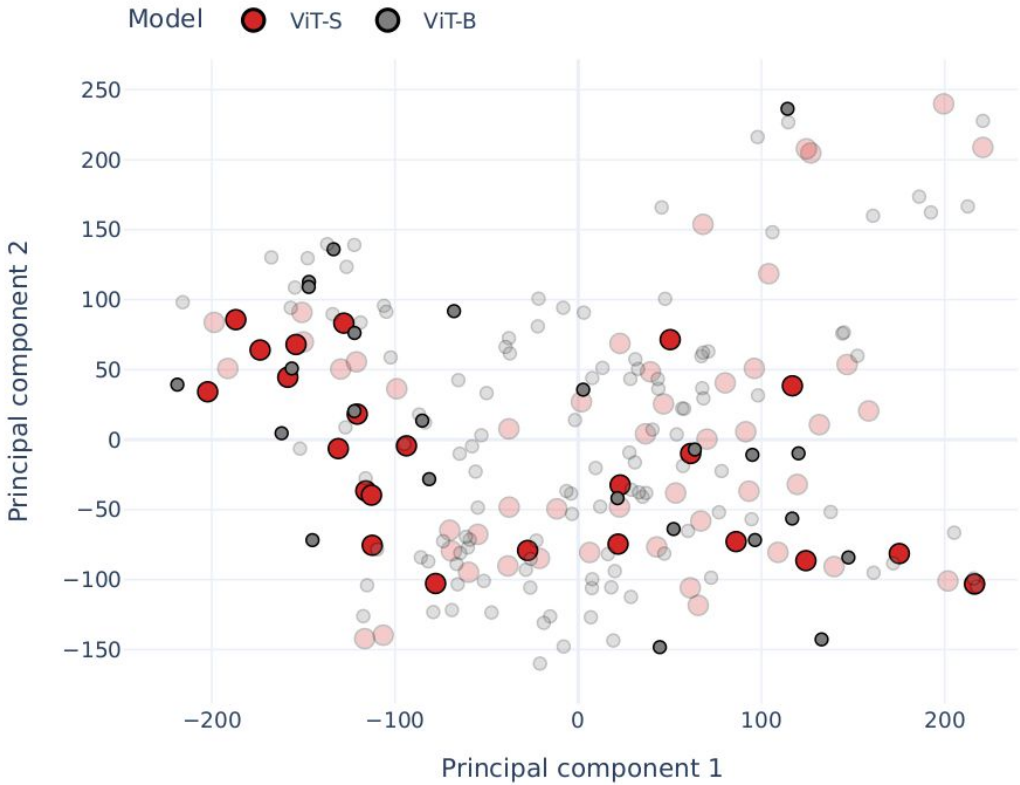
- Let $\mathbf{P} \in \mathbb{R}^{\mathrm{H} \times 3\mathrm{N}}$ be a matrix whose rows correspond to attention heads and columns the maximum, average and standard deviation of individual patch tokens for *N* images.
- We reduce the size of feature vectors from *3N* to 2.



Principal component analysis – heads per layer



Principal component analysis – heads retained after pruning



Conclusions



- Analogous attention heads operate not only within contiguous blocks, but also across varying depths of the network.
- We distinguish between “convolutional” heads and those responsible for extracting complex semantic dependencies.
- We **differentiably** select a subset of important heads.
- We maintained competitive classification accuracy while **significantly reducing FLOPs**.

What's next?

We consider a few directions as potential follow-ups:

- More aggressive pruning;
- Application of DAVE to generate Laplace maps with higher resolution;
- Analysis of attention heads with Leiden communities;
- VLM models, modification of FixLIP;
- Steering task;
- Agent-to-agent explanations...

Thank you for listening!